

WAZUH Tutorial: Introduction, Setup, and Example Use Cases

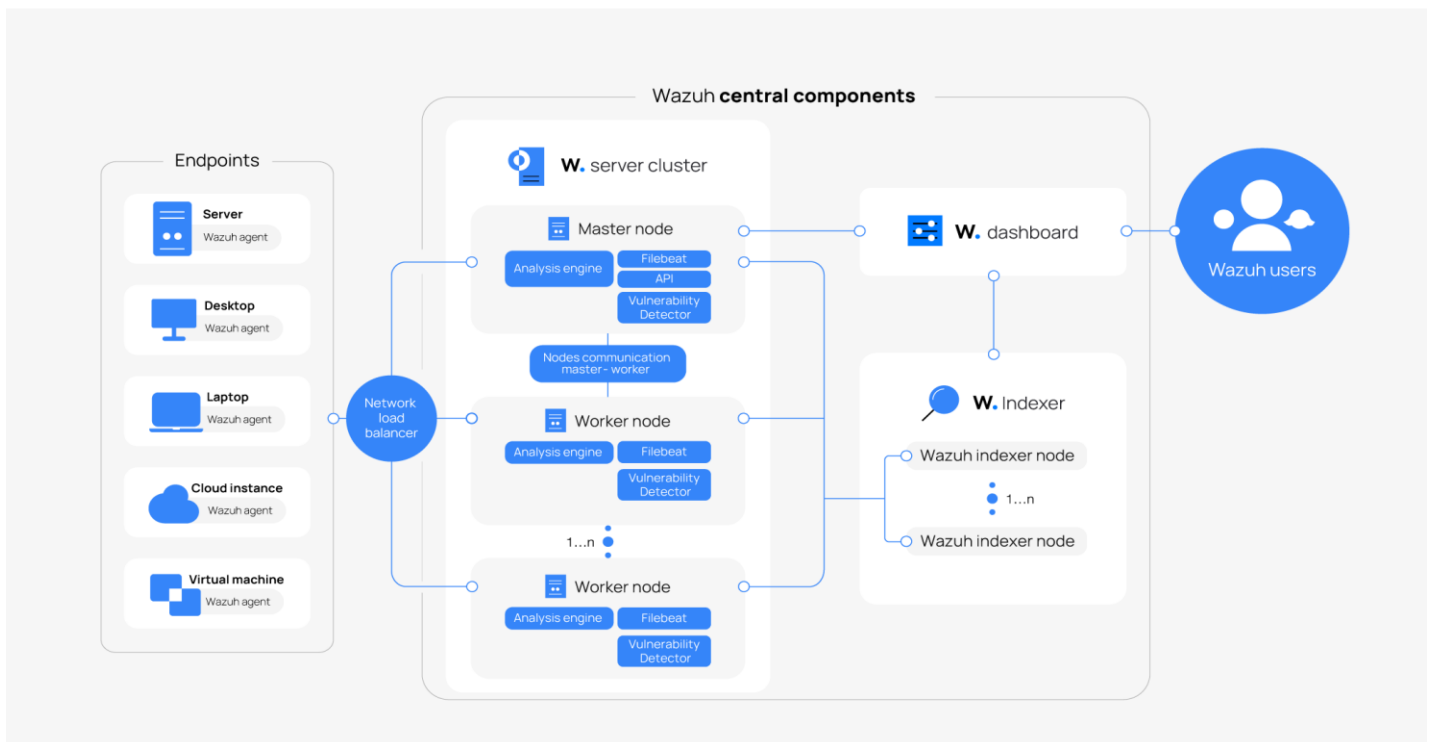
Wazuh is a free and open-source security platform that provides unified Security Information and Event Management (SIEM), threat detection, incident response, and compliance monitoring.

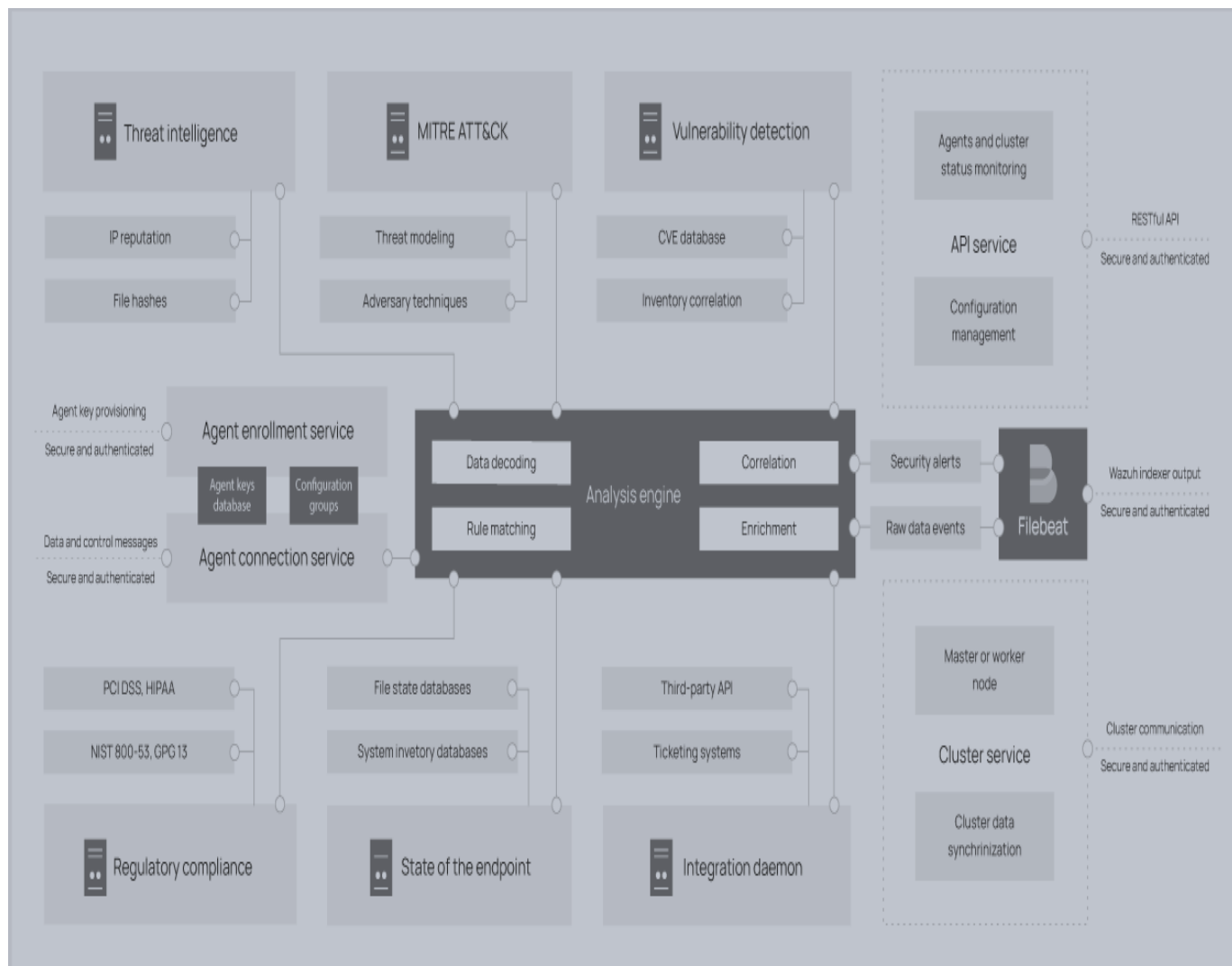
It helps monitor:

- Log data from endpoints
- File integrity changes
- Vulnerabilities and malware
- Network-based attacks

Wazuh is composed of three main components:

1. **Wazuh Manager** – collects, processes, and analyzes security data
2. **Wazuh Agent** – installed on monitored endpoints to collect logs
3. **Wazuh Dashboard** – web interface (based on Kibana) to view alerts and reports





Wazuh Use Cases

1. **Intrusion Detection:** Detects suspicious login attempts, brute force, or command injections
2. **Log Collection and Analysis:** Parses logs from Apache, SSH, Suricata, etc.
3. **File Integrity Monitoring (FIM):** Alerts when sensitive files are modified
4. **Regulatory Compliance:** Supports PCI-DSS, HIPAA, GDPR, NIST, and more
5. **Threat Intelligence Integration:** Enrich alerts with threat feeds (OTX, VirusTotal)

Wazuh Architecture and Core Components

Wazuh operates as a modular and scalable security platform composed of several key components, each responsible for different parts of the detection and monitoring lifecycle.

1. Wazuh Manager

- Central brain of the Wazuh platform.
- Receives and processes data from agents.
- Applies rules and decoders to generate alerts.
- Sends parsed alerts to the indexer.

Example: When a Wazuh agent sends SSH login activity, the manager uses decoders to parse the log and rules to identify brute force patterns. If 5 failed logins occur in 2 minutes, the manager raises an alert.

2. Wazuh Agent

- Lightweight software installed on monitored endpoints (Linux, Windows, macOS).
- Collects system logs, monitors files, checks for vulnerabilities.
- Forwards this data to the manager for correlation.

Example: On an Ubuntu server, the Wazuh agent detects changes in `/etc/passwd` and forwards a syscheck alert to the manager.

3. Wazuh Indexer (formerly Elasticsearch)

- Stores and indexes alerts received from the manager.
- Enables fast searching, aggregation, and analytics.
- Powers the data backend for the dashboard.

Example: A security analyst runs a query for "CVE" and gets all systems affected by critical vulnerabilities reported by the manager.

4. Wazuh Dashboard (Kibana-based UI)

- Web-based interface to visualize alerts, system status, compliance, etc.
- Includes modules for threat hunting, vulnerability view, file integrity, PCI-DSS, and more.

Example: An analyst sees a real-time alert labeled PCI-DSS 10.2.5 indicating an unauthorized file access event on a monitored system.

5. Filebeat (Optional)

- Ships manager logs to the indexer for ingestion.
 - Useful in distributed environments.
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Functionalities of Wazuh (With Examples)

1. File Integrity Monitoring (FIM)

Detects changes to critical files (e.g., /etc/passwd)

Example:

- Agent monitors /etc
- Admin modifies /etc/hosts
- Wazuh generates an alert labeled syscheck with severity level

2. User and Rootkit s and checks for rootkits using its active response and syscollector modules.

Example:

- Unusual login at 3AM from an unknown user triggers sshd alert
- Dashboard shows login IP, user, time

3. Intrusion Detection and Log Monitoring

Integrates with applications like Apache, SSH, and Suricata.

Example:

- Suricata logs are parsed by Wazuh agent from /var/log/suricata/eve.json
- Alerts show signatures like "SYN Flood Detected"

4. Vulnerability Detection

Using the vulnerability detector module, Wazuh correlates OS/software versions against known CVEs.

Example:

- Outdated OpenSSL detected

- Alert shows CVE-ID and severity (e.g., CVE-2023-9999, severity 9.8)

5. Regulatory Compliance

Wazuh supports mapping alerts and configurations to frameworks like:

- PCI-DSS
- HIPAA
- GDPR
- NIST 800-53

Example:

- PCI-DSS compliance dashboard highlights password policy violations
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Step 1: Download and install the agent on the client machine

```
curl -sO https://packages.wazuh.com/4.7/wazuh-agent-4.7.2.deb
```

```
sudo dpkg -i wazuh-agent-4.7.2.deb
```

Step 2: Register the agent with the manager

```
sudo /var/ossec/bin/agent-auth -m <WAZUH-MANAGER-IP> -A <agent-name>
```

Step 3: Start the agent

```
sudo systemctl enable wazuh-agent
```

```
sudo systemctl start wazuh-agent
```

Example: Detecting Unauthorized File Changes

1. On the client system (agent), enable file integrity monitoring in:

```
sudo nano /var/ossec/etc/ossec.conf
```

Add inside <ossec_config>:

```
<syscheck>
```

```
  <directories check_all="yes">/etc,/home</directories>
```

```
</syscheck>
```

2. Restart the agent:

```
sudo systemctl restart wazuh-agent
```

3. Modify a file:

```
echo "unauthorized" >> /etc/passwd
```

4. Check alerts in Wazuh dashboard → **Security Events** → **syscheck**
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Visualizing Alerts

- Go to **Threat Hunting**
- Use filters like:

```
data.alert.level:[8 TO 15]
```

- View categories like:
 - malware
 - sshd
 - suricata
 - syscheck
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Additional Functional Modules in Wazuh (Advanced Examples)

6. Log Data Collection and Custom Parsing

Wazuh uses **decoders** and **rules** to parse custom logs from any source.

Example:

- You configure Wazuh to monitor `/var/log/nginx/access.log`
- Wazuh parses failed HTTP requests and generates alerts for 403/404 codes

7. Active Response

Wazuh can automatically react to suspicious behavior by executing scripts (e.g., block IP).

Example:

- Too many failed SSH logins from 192.168.1.100

- Wazuh executes a firewall command to drop that IP via active response

8. MITRE ATT&CK Mapping

Wazuh maps many of its detection rules to MITRE ATT&CK tactics and techniques.

Example:

- A rule matches brute-force activity → Wazuh assigns it T1110 (Credential Access)
- Viewable in Dashboard > ATT&CK Matrix

9. Cloud and Container Monitoring

Wazuh can monitor Docker, Kubernetes, and cloud services like AWS using agentless methods or cloud APIs.

Example:

- Wazuh ingests AWS CloudTrail logs and alerts on IAM privilege escalations

10. Centralized Configuration and Agent Management

From the Wazuh dashboard, you can:

- Remotely restart agents
- Push new configuration files
- Trigger integrity checks

Example: You apply a centralized syscheck configuration to 50 Linux servers without logging into each one